

LAB 4 MOSFET Gate Capacitance (AC Analysis)

Technology: 22nm PTM-HP

Nominal VDD = 0.8 V

$k = 1.3 (W_p/W_n)$, $W_n = 44\text{nm}$, $W_p = k \cdot W_n$ (from Lab 1)

Single PMOS or single NMOS device wired as a MOS capacitor: drain, and source tied together to either VDD or GND, with the gate driven by VGATE (DC bias swept 0.2-0.8 V, AC magnitude 1 mV). Total gate capacitance C_{gg} is extracted from the AC gate current at each DC bias point.

Lab 4 Part A

Gate capacitance extracted vs. VGATE

$$C_{gg} = -I_i / (2 \cdot \pi \cdot f \cdot V_{ac})$$

VGATE (V)	PMOS, S/D/B @ VDD (aF)	PMOS, S/D/B @ GND (aF)	NMOS, S/D/B @ GND (aF)	NMOS, S/D/B @ VDD (aF)
0.2	55.105	30.096	31.376	18.524
0.4	45.690	26.705	32.854	19.445
0.6	43.249	25.406	39.855	21.877
0.8	37.990	24.968	41.736	26.846

Lab 4 Part B

Gate capacitance from the model card

$$C_{ox} = \epsilon_{srox} * \epsilon_{s0} / t_{oxe} \quad (\text{F/m}^2, \text{ oxide capacitance per unit area})$$

$$C_{gg} = C_{ox} * W * L \quad (\text{F}, \text{ total gate capacitance})$$

$\epsilon_{s0} = 8.854\text{e-}12 \text{ F/m}$ (vacuum permittivity), $\epsilon_{srox} = 3.9$ (both devices, same high-k/metal-gate stack)

Device	t _{oxe} (nm)	C per area (fF/um ²)	W (nm)	L (nm)	C _{gg} (aF)
NMOS	1.05	32.886	44.0	22.0	31.834
PMOS	1.1	31.391	57.2	22.0	39.503

Comparison to a typical MIM capacitor

$$C_{MIM} = \epsilon_{srox} * \epsilon_{s0} / t_{MIM} = 7 * 8.854\text{e-}12 \text{ F/m} / 30\text{e-}9 \text{ m} = 2.065\text{e-}3 \text{ F/m}^2 = 2.065 \text{ fF/um}^2$$

Capacitor	Dielectric thickness	epsrox	C per area (fF/um ²)
Typical MIM capacitor	~30 nm	7	2.065

NMOS gate oxide capacitance per unit area (32.886 fF/um²) is about **15x** higher than this typical MIM capacitor.

Lab 4 Part C

Inverter gate capacitance

$$C_{inv} = C_{gg,NMOS} + C_{gg,PMOS} = 31.834 \text{ aF} + 39.503 \text{ aF} = 71.337 \text{ aF}$$

Component	C _{gg} (aF)
NMOS	31.834
PMOS	39.503
Inverter total	71.337